

# Accepted Service Adaptation: Continuation Prices, Transfer Coordinates, and Certified Gains

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**Abstract.** A service provider gains from adaptation only if each promised continuation remains acceptable. We study expansion of an optimized restricted contract under continuation participation, payment commitments, and switching costs. A restriction-release theorem identifies the marginal gain, its reduced curvature, and the continuation-slack bottleneck that changes the optimal policy path. Reoptimizing both classes shows why greater continuation capacity can reduce the adaptation premium while greater switching friction can increase it. A second result constructs continuous promised-payment policies and global value bounds without enumerating public histories. Deterministic mixtures preserve exact promises and participation; recursive continuation prices certify an explicit error envelope, including at boundary promises. We quantify the value lost by omitting endogenous state and give exact examples separating promised payment from inherited service tiers. Common-instance comparisons distinguish fixed, scheduled, regime-dependent, limited-memory, and full-history contracts. Continuous-state experiments report verified uniform error bounds. Correlation-preserving cached prices certify joint changes in commitments and costs, including uniform bounds over parameter cells. Earlier unfavorable learned-implementation comparisons are retained. The results identify which contractual information creates value and how that value can be implemented with verifiable continuation guarantees.

**Keywords:** continuation participation; service adaptation; contractual flexibility.

**Subject classifications:** Dynamic programming/optimal control: service adaptation; Inventory/production: service contracts; Programming: parametric convex optimization.

**Area of review:** Stochastic Models.

# 1 Introduction

A service provider can find a more profitable operating policy and still be unable to offer it. A customer with an exit right evaluates the promised service and transfers remaining at the current review, not only the surplus forecast when the agreement was signed. An attractive aggregate adjustment can therefore fail in a particular continuation. An operational improvement must respect these continuation commitments as well as physical capacity and the cost of changing service tiers.

The relevant comparison is with the best implementable restricted contract. A time-only amendment, for example, may be appropriate when a service organization can publish a calendar of adjustments but cannot maintain a separate rule for every observed regime. A current-regime amendment permits more information. A limited-memory amendment permits more still. Comparing an adaptive policy with an arbitrary fixed protocol would mix the value of additional flexibility with the cost of having selected the original protocol poorly. We optimize both sides of each comparison.

This distinction changes the comparative statics. Relaxing a continuation cap improves each optimized value, but it need not improve their difference. A restricted contract may benefit more from the relaxation than an already adaptive one. Conversely, a restriction can force costly changes that the full class can avoid; increasing switching friction can then increase the value of removing the restriction. We give exact examples, rather than infer the adaptation premium from a comparative static for only one of the two values.

Our main structural result describes a controlled release of contractual restrictions. Starting from an optimized restricted contract, it identifies the marginal value of permitting unequal decisions at previously pooled histories, the curvature of this value, and the first continuation constraint that exhausts its slack. The result gives an exact policy path until the first change in the active contractual constraints. Successive segments describe the complete release frontier. It also gives the smallest release needed to attain a specified gain on the initial segment. The underlying parametric optimization tools are established; the result identifies the operational objects that must be priced and the continuation bottlenecks that limit an accepted improvement.

A second contribution is the comparator-adjusted rent decomposition. It attributes changes in the adaptation premium to the difference between full-class and restricted-class continuation prices. It separates the value of adding information from the value of relaxing participation or capacity. A hierarchy of optimized contracts then assigns each increment of value to a specified expansion of the implementable menu. These increments are conditional on the ordering of the restrictions, not an invariant allocation of value among information features.

We make the exit-review institution primary. Public demand, fulfillment, service tiers, and the transfer schedule are contractible; there is no hidden effort or private type. A promised-payment state summarizes the remaining participation obligation when the primitives and outside opportunities have a finite Markov description. This is an augmented state: observing the current demand regime alone does not reveal the inherited tier or the remaining payment promise. The recursion is exact under its stated sufficiency conditions. It does not turn an exponentially growing history tree into a universally polynomial long-horizon problem. We therefore accompany the continuous-state recursion with feasible policy mixtures, recursive price upper bounds, and explicit error certificates over the entire promise domain. Exact examples and a curvature bound

identify the value lost when a smaller table omits either endogenous coordinate.

Finally, a policy certificate and an economic certificate answer different questions. The first checks whether an implemented decision is accepted and sufficiently close to the full optimum. The second checks whether its value exceeds an optimized restriction. We provide reusable upper envelopes for the latter task both when only rewards vary and when the same shock changes rewards, continuation coefficients, budgets, and switching costs. Retaining these correlated coefficients avoids the need to replace actual commitments by a common relaxed set. On a retained priced face, the exact-anchor reuse loss is quadratic; cellwise bounds also cover a continuous uncertainty set. We also explain why the robust comparator used in the earlier computational record converges at small uncertainty radii despite a steep initial slack curve. These results support deployment accounting; they do not replace the contractual contribution with a claim about a particular prediction architecture.

The numerical study follows this hierarchy. On the same legacy instances we compare fixed, time-only, regime-dependent, two-period-memory, and full-history amendments. A separate public-Markov family compares actual service tiers rather than amendments to heterogeneous legacy tiers. All comparisons retain the same accepted set and use independently recomputed rational bounds. We retain the earlier finding that the tested learned routes do not amortize against lifted classical continuation under the matched certified protocol. No computational advantage for learning is needed for the structural value results.

## 2 Relation to Dynamic Contracting and Optimization

Continuation states are established in dynamic contracting. Spear and Srivastava (1987) and Thomas and Worrall (1988) use recursive contractual promises in environments with incentive or enforcement constraints. In Operations Research, Chen et al. (2020) study optimal monitoring when unobserved effort changes the arrival of adverse events. Their formulation uses the agent’s promised continuation utility as a state, with promise keeping, incentive compatibility, and continued participation determining its evolution. Liang et al. (2023) study resource-allocation contracts that induce effort to reduce adverse events and derive implementable allocation structures using dynamic incentives and self-generating sets.

Our public-state institution does not solve those information problems. Its controls are contingent service tiers under externally fixed prices, and its binding constraints are continuation participation and physical or payment commitments. The question is which accepted deviations improve upon an already optimized information restriction. A *continuation price* here is a multiplier on a continuation participation constraint. It is not the customer’s promised utility. The promised payment in Section 6 is instead a primal state variable. This distinction prevents a dual representation from being mistaken for a new promised-utility construction.

The journal-local boundary also includes maintenance, capacity partnerships, and dynamic assignment. Tian et al. (2021) choose payments and potential termination to induce unobserved maintenance and repair effort. Liu et al. (2019) design renegotiated membership agreements and transfers under privately updated demand information. He (2023) derives deadline mechanisms for project assignment with a privately evolving cost type and limited capacity. Here demand and service are public, prices are fixed, and customer exit is a participation constraint rather than a privately informed report or an endogenous termination instrument.

Switching friction enters the provider's operating objective. The structural question is the value of relaxing a specified public-information restriction, and its implementation through an exact remaining-payment state; we do not claim a solution to those incentive-design problems.

Constrained Markov decision processes supply resource-state and occupation-measure methods (Altman, 1999). Strong convexity, normal cones, and sensitivity of convex values are standard (Boyd and Vandenberghe, 2004). Absolute-value switching is an instance of a generalized-lasso penalty (Tibshirani and Taylor, 2011). Explicit parametric quadratic programming supplies the piecewise-affine policy and piecewise-quadratic value structure used below (Bemporad et al., 2002). We do not claim any of these general principles as new. Our use of them distinguishes restriction rents from participation rents, identifies the continuation-slack event that terminates a profitable release, and accounts for reoptimization of the comparator. The transfer identity by itself establishes none of those value comparisons.

Robust set containment is also an established device (Ben-Tal and Nemirovski, 2000; Bertsimas and Sim, 2004). We use it only for certification and quantify its local loss. Numerical optimizers, including OSQP (Stellato et al., 2020), produce candidate decisions and prices. The evidence below accepts a decision only after a separate rational calculation has checked feasibility and its primal–dual bracket. The historical value-gradient and learned-price results remain in the accompanying research archive, with their assumptions and proofs intact; they are implementation results rather than the identity of this paper.

### 3 A Service Agreement with Continuation Participation

#### 3.1 Primitives and accepted continuations

At date  $t = 0, \dots, T - 1$ , a provider observes a public regime  $z_t$  and an inherited tier  $q_t$ . It chooses stock  $S_t$  and a tier  $x_t \in [0, 1]$  before demand is realized. Demand  $D_z$  is exogenous, nonnegative, and has finite mean. Regime transitions do not depend on the policy. The next inherited tier equals the chosen tier. Premium  $r_z x$ , indemnity  $\kappa x(D_z - S)^+$ , and physical cost parameters are fixed in the agreement. There is no private information. With  $f_z(S) = \mathbb{E} \min(S, D_z)$  and  $L_z(S) = \mathbb{E}(D_z - S)^+$ , the customer's expected stage utility is

$$u(z, S, x) = \nu f_z(S) - a(z, S)x, \quad a(z, S) = r_z - \kappa L_z(S), \quad \nu > 0. \quad (1)$$

Maintenance and switching costs are resources expended by the provider, not transfers received by the customer.

An operational interpretation is a service-level agreement or maintenance subscription with verifiable fulfillment and periodic cancellation rights, rather than an unrestricted mechanism-design problem. The customer may leave *before implementation at every review*. Remaining contractual utility must exceed the publicly specified outside continuation  $\underline{U}(h_t)$  at every positive-probability history  $h_t$ . The provider commits to its announced contingent protocol; the exit right does not let it retrospectively change prices or tiers. We consider contracts intended to retain the customer at all modeled reviews. A model in which termination is itself an optimized action would need an additional termination decision.

Stock can be contractually specified. It can also be identified from service and cost commitments: if  $s_z$

uniquely minimizes  $C_z(S) - \xi f_z(S)$ ,  $\xi > 0$ , then  $F^\pi \geq F^s$  and  $C^\pi \leq C^s$  force  $S_t = s_{z_t}$  at every reachable review. The nonnegative expected sum of the minimization gaps must be zero. EC Section EC.1 gives the boundary, vector, and relaxed-commitment statements. Thus the tier reduction need not assume away a stock decision that the accepted physical commitments already determine. Without such identification, stock must remain in the accepted optimization.

On a finite exogenous tree, let  $w_n = \beta^{t(n)} \Pr(n) > 0$ ,  $c_n = w_n a(z_n, s_{z_n})$ , and  $i \succeq n$  mean that  $i$  is in the subtree rooted at  $n$ . Multiplying conditional participation by  $w_n$  gives

$$\sum_{i \succeq n} c_i x_i \leq b_n, \quad b_n = \sum_{i \succeq n} w_i v f_{z_i}(s_{z_i}) - w_n \underline{U}(n). \quad (2)$$

This derivation imposes every continuation constraint directly. Ex ante participation alone is the special case retaining only the root row. It does not imply the nonroot inequalities. A fixed root payment, when stipulated in the agreement, gives  $\sum_i c_i x_i = b_0$ ; it is a separate payment commitment, not an equality inferred from participation. General formulations below also allow a root inequality.

### 3.2 Optimized restrictions and switching

Write the accepted set, including the tier box, as

$$X_b = \{x \in C : Ax \leq b, Ex = e\}, \quad C = [\ell, u], \quad W_\lambda(x) = p^\top x - \frac{1}{2} x^\top Q x - \lambda T(x), \quad (3)$$

where  $Q \succ 0$ ,  $\lambda \geq 0$ , and

$$T(x) = \sum_{a=1}^m \kappa_a |(Dx - h)_a|, \quad \kappa_a > 0.$$

The linear reward  $p$  includes net receipts and the linear part of convex operating costs, for example a quadratic cost around a service-specific efficient tier. It need not be proportional to the payment vector  $c$ . The general quadratic model and its synthetic examples are not asserted to be estimates of the stock institution above.

Rows of  $A$  include (2) and any other retained capacity constraints.  $E$  contains fixed payment or physical equalities.  $D$  can include temporal and cross-service differences;  $h$  incorporates an installed initial tier. The vector-service case uses the same stacked notation, and signed payment coefficients are allowed in (3). Only the specialized transfer lemma requires positive scalar coefficients. All accepted sets used below are nonempty and compact. Strong concavity makes each optimum unique.

A contractual restriction is a predeclared equality  $Mx = m_0$ , and

$$Y_b = X_b \cap \{Mx = m_0\}, \quad K_\lambda(b) = \max_{x \in Y_b} W_\lambda(x), \quad J_\lambda(b) = \max_{x \in X_b} W_\lambda(x), \quad \Delta_\lambda(b) = J_\lambda(b) - K_\lambda(b). \quad (4)$$

The matrices and commitments are the same on both sides. Pooling decisions at histories with the same permitted information gives one important  $M$ . The experiments also pool *amendments*  $x - z^{\text{legacy}}$ ; then  $m_0 = Mz^{\text{legacy}}$ , which is generally different from pooling actual tiers. We keep that distinction explicit.

### 3.3 Continuation prices and relative transfers

For a comparator  $z \in X_b$ , put  $g = p - Qz$ . Convex optimality gives the following retained foundation.

**Proposition 3.1** (Accepted balance). *The comparator  $z$  is optimal over  $X_b$  if and only if there exist  $\mu \geq 0$ , unrestricted  $\eta$ ,  $v \in N_C(z)$ , and switching tensions  $s$  such that*

$$g = A^\top \mu + E^\top \eta + v + D^\top s, \quad \mu_i(A_i z - b_i) = 0, \quad (5)$$

with  $s_a = \lambda \kappa_a \text{sign}(Dz - h)_a$  on nonzero differences and  $|s_a| \leq \lambda \kappa_a$  on zero differences. If it is not optimal, a feasible sufficiently short direction has strictly positive directional gain. For optimality over  $Y_b$ , add  $M^\top \xi$  to (5);  $\xi$  prices the restriction rather than a customer participation constraint.

*Proof.* The objective loss is the sum of a differentiable convex quadratic and a finite convex absolute-value function. Its subgradient plus the polyhedral normal cone characterizes a minimum. Polyhedral normals give the displayed multipliers, without a strict-feasibility assumption. The alternative follows from the directional derivative test for a convex function on a polyhedron.  $\square$

In the scalar tree specialization, assume  $c_n > 0$  and a fixed root payment. For any feasible  $z$ , define its continuation slack  $\sigma_n = b_n - \sum_{i \geq n} c_i z_i$  at nonroot nodes. Let  $B$  have column  $n$  equal to  $-1$  at row  $n$  and  $c_n/c_{\text{pa}(n)}$  at its parent, with zeros elsewhere.

**Lemma 3.2** (Benchmark-relative transfer coordinates). *The correspondence*

$$x = z + By, \quad y_n = -c_n^{-1} \sum_{i \geq n} c_i (x_i - z_i) \quad (6)$$

is a bijection between  $X_b$  and the relative transfers satisfying  $y_n \geq -\sigma_n/c_n$ , the original tier box, and every additional inequality or equality after substitution. The restriction  $Mx = Mz$  becomes  $MBy = 0$ . Both directions of the map take linear work in the number of tree nodes.

Subtree telescoping proves the lemma; EC Section EC.2 gives the inverse explicitly. Negative transfers consume pre-existing continuation slack and cannot be discarded. The lemma is a representation, not a theorem about how much an optimized restriction costs. We next use the restriction prices and the slack in a value theorem.

## 4 The Value Frontier of Releasing a Restriction

Fix  $b, \lambda$  and the optimized comparator  $z = \arg \max_{Y_b} W_\lambda$ . Let  $\omega \in \mathbb{R}_{++}^k$  set the units of permitted deviations from the restriction. Define

$$V(t) = \max\{W_\lambda(x) : x \in X_b, |Mx - m_0| \leq t\omega\}, \quad t \geq 0. \quad (7)$$

Thus  $V(0) = K_\lambda(b)$ . The release does not relax participation: it permits previously pooled decisions to differ while preserving every accepted constraint. The full value is reached for finite  $t$ , because  $X_b$  is compact. Scaling a row of  $M$  requires the same scaling of its  $\omega$ ; otherwise it changes the economic release unit.

#### 4.1 An exact release segment and its first bottleneck

To state the explicit segment, fix a regular active cell at  $z$ . Collect permanent equalities, active accepted inequalities and tier bounds, and zero switching differences into  $Rx = r$ . Retain the signs of all nonzero switching differences and absorb their linear cost into  $\bar{p}$ . Let  $R$  and  $[R^\top, M^\top]^\top$  have full row rank. Suppose the active accepted-inequality multipliers are strictly positive, zero-edge tensions are strictly within their capacities, inactive constraints and nonzero switching signs have positive clearance, and the restriction multiplier  $\xi$  has no zero coordinate. These are local regularity conditions, not assumptions of the global frontier in the theorem. Rows with zero switching coefficient need not be included as kink constraints.

Choose any matrix  $Z$  whose columns form a basis for  $\ker R$ , and set

$$H = Z^\top QZ, \quad S = [MZH^{-1}Z^\top M^\top]^{-1}, \quad P = ZH^{-1}Z^\top M^\top S, \quad d = Pv, \quad v_j = \omega_j \text{sign } \xi_j. \quad (8)$$

These objects are independent of the choice of basis.  $S$  is the marginal curvature of releasing the restriction after the currently binding accepted constraints have been enforced.

**Theorem 4.1** (Restriction-release frontier). *For (3)–(7),  $V$  is continuous, nondecreasing, concave, and piecewise quadratic. Its unique tier policy is continuous and piecewise affine. At almost every  $t > 0$ ,*

$$V'(t) = \omega^\top (\xi^+(t) + \xi^-(t)), \quad V(t) - V(0) = \int_0^t \omega^\top (\xi^+(s) + \xi^-(s)) ds, \quad (9)$$

where  $\xi^\pm$  are optimal multipliers on the two release bounds. At differentiability points the displayed directional price is independent of the choice of optimal multipliers.

Under the regular-cell conditions, let  $t_1$  be the first nonnegative time at which an inactive accepted constraint or tier bound binds, a nonzero switching difference reaches zero, an old active-inequality multiplier reaches zero, a zero-edge tension reaches a capacity, or a component of  $\xi - tSv$  reaches zero in its original signed direction. Then  $t_1 > 0$  and, for  $0 \leq t \leq t_1$ ,

$$x(t) = z + td, \quad V(t) - K_\lambda(b) = t \sum_j \omega_j |\xi_j| - \frac{1}{2} t^2 v^\top S v. \quad (10)$$

Each event time is an explicit linear ratio obtained from the primal and dual paths. In particular, a currently slack continuation row can be the first bottleneck only at

$$t_n = \frac{\sigma_n}{\sum_{i \geq n} c_i d_i} \quad \text{when } \sum_{i \geq n} c_i d_i > 0; \quad (11)$$

otherwise that row consumes no slack along this segment. In transfer coordinates this is exactly the time at which its lower bound  $y_n \geq -\sigma_n/c_n$  is reached.

*Proof.* Mixing policies feasible at two release levels proves concavity; nesting proves monotonicity. Introduce epigraphs for the absolute values. For fixed active inequalities and switching signs the KKT equations are linear in the parameter, so the unique tier solution is affine and its value is quadratic on that cell. There are finitely many active patterns. Dependent equations may be removed and a finite refinement made; no nondegeneracy assumption is needed for this global piecewise description. Compactness and uniqueness give continuity. Optimal multipliers support the concave value and give (9); integration is valid for this continuous piecewise-quadratic function.

For the explicit segment, a prescribed restriction displacement  $\delta$  has the cell optimum  $x = z + P\delta$ . Solving the equality-constrained quadratic and using  $RP = 0$ ,  $MP = I$ , and  $P^\top QP = S$  gives the gain  $\xi^\top \delta - \delta^\top S\delta/2$ . Its maximum over  $|\delta| \leq t\omega$  is at  $\delta = tv$  while  $\xi - tSv$  retains its signed direction. The old active multipliers and kink tensions also evolve affinely. Until one of the listed events, their signs and capacities, the remaining primal inequalities, and the nonzero switching signs all remain valid. They therefore supply the *full* KKT certificate, not just a solution restricted to the cell. Continuity includes the first event. Substitution into a subtree inequality gives (11). The detailed dual path and degeneracy qualifications are in EC Section EC.3.  $\square$

Write  $a = \sum_j \omega_j |\xi_j|$  and  $q = v^\top Sv > 0$ . For a desired gain  $g \in [0, V(t_1) - K]$ , the smallest release on this initial segment is

$$t(g) = \frac{a - \sqrt{a^2 - 2qg}}{q}. \quad (12)$$

No accepted improvement can attain this gain with a smaller release, because (10) is the optimized value, not the gain of a selected direction. At later events one changes the active cell and applies the same equality-constrained calculation. The number of cells need not be polynomial; the theorem is an exact structural path, not a universal complexity claim. If a regularity condition fails, (9) and the global frontier remain valid, but the signed-corner formula is replaced by the appropriate small release quadratic program or a degenerate active-set refinement.

## 4.2 A complete three-node frontier

Consider a root and two children,  $c_i = 1$ , tiers in  $[0, 1]$ , root payment  $x_0 + x_1 + x_2 = 3/2$ , and child caps  $x_1, x_2 \leq 3/4$ . Let

$$W(x) = x_1 - \frac{1}{2}(x_0^2 + x_1^2 + x_2^2), \quad Y = \{x \in X : x_1 = x_2\}.$$

The optimized restriction is  $z = (1/6, 2/3, 2/3)$  with value  $5/24$ . Releasing  $|x_1 - x_2| \leq t$  gives

$$V(t) - K = \begin{cases} t/2 - t^2/4, & 0 \leq t \leq 1/6, \\ -1/48 + 3t/4 - t^2, & 1/6 \leq t \leq 3/8, \\ 23/192, & t \geq 3/8. \end{cases} \quad (13)$$

The first policy segment is  $(1/6, 2/3 + t/2, 2/3 - t/2)$ . Child 1 has slack  $1/12$ , consumes it at rate  $1/2$ , and therefore binds at  $t = 1/6$ . The second segment is  $(t, 3/4, 3/4 - t)$ . At  $t = 3/8$ , the release price becomes

zero and the full optimum  $(3/8, 3/4, 3/8)$  is reached. Its relative transfer is  $(-1/12, 7/24)$ ; the negative coordinate uses exactly the initially available child-1 slack. Thus the continuation slack determines an actual change in the optimal policy path, rather than only the domain of a coordinate map. EC Section EC.4 supplies exact primal and dual witnesses for both segments.

## 5 Capacity, Friction, and Optimized Comparator Rents

### 5.1 Relaxing a continuation can reduce the adaptation premium

Let  $b(t) = b^0 + td$ , with  $d \geq 0$ , and suppose  $Y_{b(t)}$  is nonempty on the interval considered. Both optima in (4) are reoptimized for each  $t$ .

**Theorem 5.1** (Comparator-adjusted continuation rents). *The full and restricted values are continuous, nondecreasing, concave, and piecewise quadratic along the budget path. If  $\mu_X(t)$  and  $\mu_Y(t)$  are optimal accepted-inequality multipliers, then, at almost every  $t$ ,*

$$\frac{d}{dt}\Delta_\lambda(b(t)) = d^\top(\mu_X(t) - \mu_Y(t)), \quad \Delta_\lambda(b(v)) - \Delta_\lambda(b(u)) = \int_u^v d^\top(\mu_X - \mu_Y) dt. \quad (14)$$

*Consequently, the adaptation premium is nondecreasing along the path if and only if the rent difference is nonnegative almost everywhere. At a degenerate point, the right directional value derivative along  $d$  is the minimum of  $d^\top \mu$  over that problem's optimal multiplier set; one must compute the two directional derivatives separately, rather than subtract arbitrary selected multipliers.*

*Proof.* Mixing and nesting give the first two value properties. Fixed-matrix parametric quadratic programming gives the piecewise description as in Theorem 4.1. Weak duality makes each accepted multiplier a supporting price for the budget. At differentiability points all such directional prices coincide with the derivative. Subtract and integrate the two value derivatives. A continuous piecewise-quadratic function is nondecreasing precisely when its derivative is nonnegative almost everywhere. The one-sided rule follows by taking the directional derivative of the infimum of affine-in-budget dual objectives; EC Section EC.5 supplies the polyhedral qualification.  $\square$

An exact example makes the distinction material. Keep the preceding three-node objective and root payment, but set  $x_1 \leq b$ ,  $x_2 \leq 3/4$ , with  $b \in [1/4, 2/3]$ . The time-only restriction remains  $x_1 = x_2$ . The full and restricted optima are

$$x^X(b) = ((3/2 - b)/2, b, (3/2 - b)/2), \quad x^Y(b) = (3/2 - 2b, b, b),$$

with cap prices  $\mu_X = 7/4 - 3b/2$  and  $\mu_Y = 4 - 6b$ . Both values increase, but

$$\Delta(b) = \frac{9}{16}(2b - 1)^2, \quad \Delta'(b) = \frac{9}{4}(2b - 1). \quad (15)$$

The premium falls to zero as  $b$  rises to  $1/2$ , then rises again. A claim that more participation slack necessarily

makes adaptation more valuable would get the sign wrong on half this interval. The relevant scarcity measure is the difference of the optimized contractual rents.

## 5.2 Friction thresholds when the restriction is reoptimized

For fixed  $X, Y$ , let  $x_\lambda$  and  $z_\lambda$  be their optima. A fixed-comparator balance test is not enough when  $z_\lambda$  changes with friction.

**Proposition 5.2** (Optimized-comparator friction path). *The adaptation premium is continuously differentiable for  $\lambda > 0$ , with a right derivative at zero, and*

$$\Delta'(\lambda) = T(z_\lambda) - T(x_\lambda). \quad (16)$$

*On every bounded friction interval its zero set is a finite union of closed intervals and isolated points. It can be computed from the restricted parametric policy path without solving a full-class optimization at every friction: refine that path at changes of its accepted active rows and switching signs, and on each resulting cell impose (5) at the affine  $z_\lambda$ . This is a linear feasibility problem jointly in  $\lambda$  and the full-class prices; its projection on  $\lambda$  is an interval, possibly empty or a point. The union of these projections is exactly the zero set. There is no general upward-ray conclusion for arbitrary restrictions.*

*Proof.* Uniqueness and compactness imply continuous optimizers. The envelope argument gives  $J'(\lambda) = -T(x_\lambda)$  and  $K'(\lambda) = -T(z_\lambda)$ , including active-set changes. Both paths are piecewise affine. On a refined cell, every sign in (5) is fixed or zero,  $z_\lambda$  is affine, and tension bounds and stationarity are linear in the displayed variables. The full balance test is necessary and sufficient. A projection of a polyhedron onto a line is an interval. Finitely many cells establish the assertion; the method does not bound their number polynomially.  $\square$

For example, use the root payment  $3/2$ , the box and child caps above,  $p = (1/2, 3/4, 1/4)$ ,  $Q = I$ , and  $T(x) = |x_1 - x_0| + |x_2 - x_0|$ . A restriction  $x_1 - x_2 = 1/2$  has optimized policy  $z = p$  for every  $\lambda$ : its switching cost is already the smallest possible under that restriction. The full optimum is  $(1/2, 3/4 - \lambda, 1/4 + \lambda)$  until  $\lambda = 1/4$ , and is constant thereafter. Therefore

$$\Delta(\lambda) = \begin{cases} \lambda^2, & 0 \leq \lambda \leq 1/4, \\ \lambda/2 - 1/16, & \lambda \geq 1/4. \end{cases} \quad (17)$$

More friction increases the value of removing a restriction that forces switching. This does not contradict constant-comparator critical-friction results: their restriction and zero-switching benchmark are different.

### 5.3 Which information expansion creates the value?

Let  $Y_0 \subseteq \dots \subseteq Y_k = X$  be nonempty predeclared classes and  $K_j = \max_{Y_j} W$ . Then

$$J - K_0 = \sum_{j=1}^k (K_j - K_{j-1}), \quad K_j - K_{j-1} \geq 0. \quad (18)$$

The identity is elementary; its economic use requires each  $K_j$  to be optimized under the *same* accepted constraints. A release path between adjacent equality-defined classes prices that particular increment through Theorem 4.1. The ordering matters when information features interact; the increments are not automatically Shapley values or unique causal attributions.

The fixed class describes an unchanging amendment or tier. The time-only class describes a preannounced schedule. The current-regime class requires public regime measurement and a date–regime decision table. A two-period-memory class additionally requires retaining the preceding regime. The full-history class has no such information pooling. These are operational restrictions, not an assertion that time-only is the universal status quo. Current regime alone can be insufficient even when exogenous primitives are Markov, because the inherited tier and the outstanding promise are endogenous state variables. The next section establishes the corresponding sufficient-state formulation.

## 6 Promised-Payment Recursion and Horizon Complexity

Suppose the regime has a finite public Markov transition matrix  $P_t$ , the one-period net payment is  $a_t(z)x$ , and the reward is  $g_t(z, q, x)$ . Suppose further that each conditional payment cap  $\bar{b}_t(z)$  and the feasible actions depend only on the displayed state and date. A general history-dependent outside option does not satisfy this assumption merely because demand is Markov.

Let  $\mathcal{V}_t(z, q, b)$  denote the maximum conditional reward from date  $t$  with an *exact* remaining discounted payment promise  $b$ . Put  $\mathcal{V}_T(z, q, 0) = 0$  and  $\mathcal{V}_T(z, q, b) = -\infty$  for  $b \neq 0$ . For  $b \leq \bar{b}_t(z)$ , define

$$\begin{aligned} \mathcal{V}_t(z, q, b) &= \max_{x, \{b_{z'}\}} g_t(z, q, x) + \beta \sum_{z'} P_t(z, z') \mathcal{V}_{t+1}(z', x, b_{z'}) \\ \text{subject to} \quad &a_t(z)x + \beta \sum_{z'} P_t(z, z') b_{z'} = b, \\ &b_{z'} \leq \bar{b}_{t+1}(z') \quad \text{for positive-probability children.} \end{aligned} \quad (19)$$

Outside this cap, set the value to  $-\infty$ . A zero-probability successor imposes no obligation. For a root upper budget rather than an equality, maximize the root value also over permitted  $b$ .

**Theorem 6.1** (Exact sufficient-state implementation). *Under the stated Markov sufficiency and compact-attainment conditions, (19) has the same value as the full accepted history-tree problem with its specified root promise. There exists an optimal deterministic contract implemented using only  $(t, z, q, b)$ . Conversely, a feasible recursion policy induces all tree continuation constraints. With several additive commitments, the result holds with a vector of remaining commitments. A history-dependent cap, nonadditive global resource,*

or omitted physical state must be included in the state or treated on the full tree; otherwise the claimed equivalence need not hold.

*Proof.* For any history policy, take its conditional remaining payments in each child subtree as the child promises. They satisfy the equality and all child caps, and induction bounds its conditional reward by the right side. Conversely, combine an attained current maximizer with child maximizers at the allocated promises. Induction gives every child obligation, while the displayed equality gives the parent's exact promise. This proves equality and constructs a state-based policy. The vector case applies the same induction to each additive commitment.  $\square$

Conditional normalization is essential. Multiplying the conditional budget at node  $n$  by  $w_n$  recovers (2). Within (19), the child weights are  $\beta P_t(z, z')$ , not unconditional probabilities multiplied a second time. Promises are decision variables allocated across next-period regimes; they are not forecasts fixed independently of the current decision.

A branching factor  $B$  and  $T$  decision dates can create  $N = (B^T - 1)/(B - 1)$  history nodes. The linear transfer map is linear in  $N$ , not in  $T$ . The recursion avoids materializing those histories when the displayed state is sufficient, but its promise state is continuous in the continuous-tier model. A grid with  $G$  promise points,  $Q_0$  inherited-tier points,  $S_0$  regimes, and  $k$  commitment coordinates has order  $TS_0Q_0G^k$  value entries; choosing child allocations is an additional cost and can grow exponentially in the number of successors. Such a grid is an approximation unless the contract institution itself has the stated finite actions and attainable promises. Our exact recursion checks use the latter finite-action interpretation, and exhaustively compare every attainable root promise with the full history enumeration. No interpolation theorem or unrestricted polynomial-horizon guarantee is inferred from them.

## 6.1 Continuous promises with executable error certificates

The sufficient-state statement is not yet an approximation guarantee. Interpolating a value table can violate an exact promise or admit a customer continuation that the contract excludes. We next construct both a feasible policy and a value bound without rounding the promise to a grid point.

Specialize to  $x, q \in [0, 1]$ ,  $a_t(z) > 0$ , and

$$g_t(z, q, x) = r_t(z)x - \frac{1}{2}d_t(z)x^2 - \ell_t(z)|x - q|, \quad d_t(z) > 0, \quad \ell_t(z) \geq 0. \quad (20)$$

Only the Markov continuation caps and the exact payment commitment are imposed in this specialization; additional nonlocal constraints require additional state. Write  $I_t(z) = [L_t(z), U_t(z)]$  for feasible remaining promises. With  $L_T = U_T = 0$ , their endpoints satisfy

$$L_t(z) = \beta \sum_j P_t(z, j) L_{t+1}(j), \quad U_t(z) = \min \left\{ \bar{b}_t(z), a_t(z) + \beta \sum_j P_t(z, j) U_{t+1}(j) \right\}. \quad (21)$$

A state is infeasible if a positive-probability successor is infeasible or the displayed lower endpoint exceeds the upper endpoint. All statements below are on feasible states;  $\mathcal{D}_t(z) = [0, 1] \times I_t(z)$  includes the boundary.

At each date and regime, retain finitely many anchors  $(q_i, b_i, l_i)$  and affine upper planes  $H_k(q, b) = A_k q + B_k b + C_k$ . An anchor includes a feasible current action and mixtures of child anchors with the exact inherited tier and remaining promise, not just a scalar value. The anchor coordinates span  $\mathcal{D}_t(z)$ . Define

$$\begin{aligned} \underline{V}_t(z, q, b) &= \max_{\alpha \geq 0} \left\{ \sum_i \alpha_i l_i : \sum_i \alpha_i = 1, \sum_i \alpha_i (q_i, b_i) = (q, b) \right\}, \\ \bar{V}_t(z, q, b) &= \min_k H_k(q, b). \end{aligned} \quad (22)$$

To generate a parent plane, choose any convex mixture of child planes for each successor  $j$ , with coefficients  $(A_j, B_j, C_j)$ . Suppress the parent date and regime in  $a, r, d, \ell, P$ . For any  $s \in [-\ell, \ell]$  and  $\eta \in \mathbb{R}$ , put

$$\begin{aligned} H(q, b) &= sq + \eta b + C, \\ C &= \psi_d \left( r - s - a\eta + \beta \sum_j P_j A_j \right) + \beta \sum_j P_j \left[ C_j + \max_{v \in I_{t+1}(j)} (B_j - \eta)v \right], \\ \psi_d(v) &= \max_{0 \leq x \leq 1} \{vx - dx^2/2\}. \end{aligned} \quad (23)$$

The maximizer in  $\psi_d$  is  $\min\{1, \max\{0, v/d\}\}$ . Thus a proposed plane has a checkable global constant even when its prices were obtained only approximately.

**Theorem 6.2** (Promise-preserving continuous-state envelopes). *Under (20) and the stated Markov assumptions, the interval recursion (21) is exact and  $\mathcal{V}_t(z, \cdot, \cdot)$  is concave on  $\mathcal{D}_t(z)$ . Start with zero terminal values and construct anchors and planes backward as above. If every anchor reward is a lower bound for its displayed feasible action and child mixtures, and every plane constant is at least (23), then*

$$\underline{V}_t(z, q, b) \leq \mathcal{V}_t(z, q, b) \leq \bar{V}_t(z, q, b) \quad \text{for every } (q, b) \in \mathcal{D}_t(z). \quad (24)$$

At a queried state, an optimizing mixture in (22), followed by its mixed action and child promises, gives a deterministic accepted policy with value at least  $\underline{V}_t$ . Repeating this procedure using only the current augmented state preserves every continuation cap and the exact root payment. Its loss is at most  $\bar{V}_t - \underline{V}_t$ .

A uniform certificate requires no unproved smoothness rate. Triangulate the anchor rectangle using stored grid anchors. For a triangle  $C_0$  with vertices  $i$ , let

$$\varepsilon_t(z) = \max_{C_0} \min_k \max_{i \in \text{vert}(C_0)} \{H_k(q_i, b_i) - l_i\}. \quad (25)$$

Then  $0 \leq \mathcal{V}_t - \underline{V}_t \leq \varepsilon_t(z)$  throughout the rectangle. The order of the minimum and maximum in (25) matters: checking only the minimum-plane gap at each vertex does not in general certify the interior.

*Proof.* The attainable parent promises are the sum of the action-payment interval and the discounted child intervals, intersected with the parent's cap. This proves (21) by induction. Deterministic convex combinations of two feasible subtree policies preserve the affine payment equalities and all continuation inequalities. Joint concavity of (20), including its inherited-tier argument, proves value concavity.

For a lower anchor, combine the displayed action with the child mixtures. Induction gives the stated reward lower bound. A convex mixture of parent witnesses remains feasible at its mixed inherited tier and promise; concavity bounds its reward from below by the mixed anchor rewards. At the next review, reoptimizing the child mixture cannot reduce that lower bound. This proves the executable lower envelope without storing a separate policy for every history.

For the upper bound, replace each child value by its chosen affine majorant, use  $-\ell|x - q| \leq -s(x - q)$ , and subtract  $\eta$  times the promise equation. Maximizing the separated action and child-promise terms gives (23). This is weak duality and requires neither an interior promise nor exact optimizer multipliers. The minimum of the resulting planes remains an upper bound.

On a triangle, barycentric interpolation of its anchor rewards is no larger than  $\underline{V}_t$ . For any one plane, the difference from that interpolation is affine and is maximized at a vertex. Minimize this bound over planes, then maximize over triangles. This proves (25).  $\square$

The construction compresses *certificates* as well as values. With at most  $A_0$  anchors and  $H_0$  planes per date–regime and at most  $B_0$  successors, dense witness storage is of order  $TS_0B_0(A_0^2 + H_0^2)$ , not the number of histories. Sparse mixtures reduce storage further. Evaluating a policy still requires a small mixture linear program; building each table requires optimization over child allocations. Neither the anchor count needed for a prescribed tolerance nor the size of rational arithmetic is bounded independently of the problem data. Concave interpolation and stochastic dual cuts are established ideas (Pereira and Pinto, 1991); the contractual conclusion here is that continuous promise interpolation can be accompanied by exact participation feasibility, recursive price witnesses, and an a posteriori policy-loss bound. No such conclusion is inferred from finite-action enumeration alone. In particular, deterministic mixing is invalid for indivisible tiers unless mixtures themselves are admissible actions.

## 6.2 What is lost by omitting part of the augmented state?

Theorem 6.1 gives zero value to additional history once the displayed state is sufficient. A smaller decision table can still lose value. That loss can be bounded from a certified full-class decision rather than attributed vaguely to “adaptivity.”

Let  $\mathcal{A}$  be the affine subspace imposing a predeclared pooling of histories, and let  $Y = X \cap \mathcal{A}$  be nonempty. For the general objective in Section 3, define  $\text{dist}_Q(x, \mathcal{A}) = \inf_{y \in \mathcal{A}} \|x - y\|_Q$ , where  $\|u\|_Q^2 = u^\top Qu$ .

**Proposition 6.3** (Certified value of omitted state). *Let  $x^*$  maximize  $W$  on  $X$ . Then*

$$J - K \geq \frac{1}{2} \text{dist}_Q(x^*, \mathcal{A})^2. \quad (26)$$

*For any accepted  $\hat{x}$  and valid  $U_X \geq J$ , put  $\epsilon = U_X - W(\hat{x})$ . The computable bound*

$$J - K \geq \frac{1}{2} \left[ \text{dist}_Q(\hat{x}, \mathcal{A}) - \sqrt{2\epsilon} \right]_+^2 \quad (27)$$

*is valid. If  $Q$  is diagonal with entries  $d_n > 0$  and  $\mathcal{A}$  only pools decisions in cells  $C$ , the squared distance is  $\sum_C \sum_{n \in C} d_n (\hat{x}_n - \bar{x}_C)^2$ , where  $\bar{x}_C = \sum_{n \in C} d_n \hat{x}_n / \sum_{n \in C} d_n$ .*

*Proof.* Strong concavity and optimality on the convex set  $X$  imply  $W(x^*) - W(y) \geq \|x^* - y\|_Q^2 / 2$  for every  $y \in X$ ; the argument also holds at switching kinks by applying strong concavity to the segment from  $x^*$  to  $y$  and letting its length tend to zero. Minimize the distance over the larger set  $\mathcal{A}$ . The same inequality gives  $\|x^* - \hat{x}\|_Q \leq \sqrt{2\epsilon}$ . Distance to a closed affine set is 1-Lipschitz in that norm. The diagonal formula is the weighted least-squares projection.  $\square$

Two exact examples isolate the omitted coordinates. In the first, a three-date tree has one root, two equiprobable date-1 regimes, and a common date-2 regime. Let  $a = \beta = 1$ ,  $x \in [0, 1]$ , stage reward  $r_n x - x^2/2$ , and reward coefficients  $(0, 1/4, -1/4, 0, 0)$  in root, left, right, left-terminal, right-terminal order. Set root promise 2, date-1 caps  $3/4$  and  $5/4$ , and terminal caps 1. The optimal accepted policy is  $(1, 1/2, 1/2, 1/4, 3/4)$ , with value  $-25/32$ . A date-regime table must pool the terminal actions and has optimum  $(1, 1/4, 3/4, 1/2, 1/2)$ , with value  $-27/32$ . The gain is  $1/16$ . The full optimizer reaches the two terminal histories with the same public regime and the same inherited tier  $1/2$ , but different remaining promises  $1/4$  and  $3/4$ . Thus inherited tier cannot substitute for the promise in these continuation decisions. The curvature lower bound is  $1/32$ .

Conversely, consider two remaining dates, one regime, the same remaining promise 1, and inherited tiers 0 and 1 occurring with equal probability. With quadratic cost  $(x^2 + (1 - x)^2)/2$ , incoming switching cost  $3|x - q|/2$ , and final switching cost  $|1 - 2x|/2$ , the informed first actions are  $1/4$  and  $3/4$ . Their average value is  $-15/16$ . A tier-blind common first action is optimally  $1/2$ , with value  $-1$ . Thus the promise does not replace the inherited tier either. EC Section EC.10 proves both examples, explains the interpolation residual bound, and specifies the independent continuous-state checks.

## 7 Implemented Gains and Reusable Comparator Certificates

A feasible implemented  $\hat{x}$  and a valid upper bound  $U_Y \geq K$  give the economic certificate

$$W(\hat{x}) - K \geq W(\hat{x}) - U_Y. \quad (28)$$

An upper bound  $U_X \geq J$  separately gives  $J - W(\hat{x}) \leq U_X - W(\hat{x})$ . The first establishes an improvement over an optimized restriction; the second establishes implementation accuracy. Neither follows from predictor error or an optimizer's termination label. EC Section EC.6 states the box-conjugate bound used by the independent verifier, retaining every participation row, equality, and switching term.

### 7.1 A comparator envelope requiring no new query solve

Consider a fixed restricted geometry  $Y$  and fixed convex cost  $\phi$  with strong-convexity matrix  $D_0 \succ 0$ :

$$K(h) = \max_{x \in Y} \{(b + U^\top h)^\top x - \phi(x)\}.$$

Let the columns of  $Z_Y$  span a linear space containing every difference of two points of  $Y$ , and choose

$$L_Y \geq \|UZ_Y(Z_Y^\top D_0 Z_Y)^{-1/2}\|^2.$$

A trace bound on the corresponding positive semidefinite matrix is sufficient and can be evaluated rationally.

**Proposition 7.1** (Certified reusable restricted-value envelope). *At an anchor  $h_i$ , let  $z_i \in Y$  and  $U_i \geq K(h_i)$  be certified, and put  $\delta_i = U_i - W_{h_i}(z_i)$ . Then, for every reward query  $h$ ,*

$$K(h) \leq U_i + (Uz_i)^\top (h - h_i) + \sqrt{2L_Y\delta_i} \|h - h_i\| + \frac{1}{2}L_Y \|h - h_i\|^2. \quad (29)$$

*The minimum of these bounds over a finite cache remains valid. After storing  $U_i$ ,  $Uz_i$ ,  $\delta_i$ , and  $h_i$ , each query uses order  $r$  arithmetic operations per anchor, where  $r$  is the reward dimension, and no restricted optimization. The bound is valid even when the anchor decision is not exactly optimal.*

*Proof.* Strong convexity gives  $\|z_i - x^*(h_i)\|_{D_0}^2 \leq 2\delta_i$ . Restricting differences to the displayed linear space bounds the gradient error by  $\|U(z_i - x^*)\| \leq \sqrt{2L_Y\delta_i}$ . The same strong-monotonicity argument gives an  $L_Y$ -Lipschitz gradient  $\nabla K(h) = Ux^*(h)$ . Apply the smooth upper Taylor bound, substitute  $U_i \geq K(h_i)$ , and bound the anchor-gradient error by Cauchy–Schwarz. EC Section EC.7 gives the full argument, including exact outward square-root rounding.  $\square$

This is a nearby-reward reuse mechanism, not a theorem for an arbitrarily changing accepted geometry. Changing continuation matrices, budgets, switching coefficients, or information restrictions requires a new valid cache or a bound that explicitly covers those changes. A loose envelope is still valid but may not certify a positive economic gain. A deployment can then request a new comparator certificate; that solve must be charged. Accepted-policy implementation itself need not perform an economic-comparator batch on every query. Theorem 7.3 supplies a separate valid bound for jointly changing constraints and friction.

## 7.2 Why the robust outer comparator is locally continuous

Write a designed family as

$$Y_\theta(\rho) = \{x \in C_0 : (A^0 + \rho A_\theta)x \leq b^0 + \rho b_\theta\},$$

where  $C_0$  is the common compact box and equality-restricted class, and  $\theta$  belongs to a compact uncertainty set. Let  $\eta_i$  bound  $\sup_{\theta, x \in C_0} |(A_\theta)_i x - (b_\theta)_i|$  and put  $O(\rho) = \{x \in C_0 : A_i^0 x \leq b_i^0 + \rho \eta_i\}$ . This radius-dependent outer class contains every  $Y_\theta(\rho)$  and equals the nominal class at zero. It is not a fixed large union bound switched on at every positive radius.

**Proposition 7.2** (Local outer-comparator loss). *Suppose an anchor in  $C_0$  has slack at least  $\gamma_i > 0$  in every perturbed row for  $0 \leq \rho \leq \rho_0$ . Suppose the objective  $W_\theta$  is uniformly  $L_W$ -Lipschitz on  $C_0$ , whose diameter is  $D_C$ . Then*

$$0 \leq \max_{O(\rho)} W_\theta - \max_{Y_\theta(\rho)} W_\theta \leq L_W D_C \max_i \frac{2\rho\eta_i}{\gamma_i + 2\rho\eta_i}. \quad (30)$$

Thus the outer-set component of certificate slack vanishes at least linearly as the radius tends to zero under these assumptions. The constant can be large when an important accepted row has little strict slack.

*Proof.* An outer-feasible decision violates any true row by at most  $2\rho\eta_i$ . Mix it with the anchor by the largest fraction in (30). Every resulting true row is feasible, and the common equalities and box remain feasible by convexity. The movement is at most that fraction times  $D_C$ . Apply the Lipschitz bound to an outer maximizer.  $\square$

The row envelope can be tightened by taking its support over the equality-restricted box rather than an unrestricted box. EC Section EC.8 also gives a valid successive-outer-set refinement. Neither a small uncertainty radius nor set containment alone establishes field robustness: the uncertainty set remains a designed, uncalibrated family. Without a uniform anchor, the linear rate above is not asserted. The finer-radius experiment distinguishes high local sensitivity from an actual positive-radius discontinuity.

### 7.3 Jointly changing commitments and switching costs

The reward-only envelope does not address the case in which the same operational shock changes revenue, continuation coefficients, capacity, and switching friction. Replacing these correlated constraints by independent row envelopes loses contractual information. We instead retain the actual coefficient vector in a reusable dual certificate.

For this construction, write  $Q = H + R^\top R$ , where  $H \succ 0$  is diagonal, and combine the payment and information equalities into  $F_\theta x = f_\theta$ . The restricted problem is

$$\begin{aligned} K_\theta &= \max_{x \in Y_\theta} W_\theta(x), & Y_\theta &= \{x \in [\ell, u] : A_\theta x \leq b_\theta, F_\theta x = f_\theta\}, \\ W_\theta(x) &= c_\theta + p_\theta^\top x - \frac{1}{2} x^\top H x - \frac{1}{2} \|R x\|^2 - \lambda_\theta \sum_j \kappa_j |(Dx - h)_j|. \end{aligned} \quad (31)$$

The common box,  $H, R, D, h$ , and positive weights  $\kappa$  are fixed; all displayed  $\theta$ -dependent coefficients may change together, with  $\lambda_\theta \geq 0$ . The known constant  $c_\theta$  permits gain-relative objectives. Each restricted class is nonempty. Define

$$\psi_i(d) = \max_{\ell_i \leq v \leq u_i} \{dv - H_{ii}v^2/2\}, \quad v_i(d) = \min\{u_i, \max\{\ell_i, d/H_{ii}\}\}.$$

A cached price  $a = (\alpha, \mu, \nu, \tau)$  has  $\mu \geq 0$  and  $|\tau_j| \leq \kappa_j$ . Its switching tension at a query is  $\lambda_\theta \tau$ , not the unscaled tension from the training problem.

**Theorem 7.3** (Correlated comparator reuse and its local loss). *For any such cached price, put*

$$d_\theta(a) = p_\theta - R^\top \alpha - A_\theta^\top \mu - F_\theta^\top \nu - \lambda_\theta D^\top \tau, \quad (32)$$

$$\mathcal{U}_\theta(a) = c_\theta + \frac{1}{2} \|\alpha\|^2 + \mu^\top b_\theta + \nu^\top f_\theta + \lambda_\theta \tau^\top h + \sum_i \psi_i(d_{\theta,i}(a)). \quad (33)$$

Then  $K_\theta \leq \mathcal{U}_\theta(a)$  for every query. The minimum over a finite cache remains valid without a new restricted optimization, an interior feasible anchor, or an unchanged active set.

More precisely, for every feasible restricted  $z$ , the certificate gap is the sum of four nonnegative terms:

$$\begin{aligned} \mathcal{U}_\theta(a) - W_\theta(z) = & \frac{1}{2} \|\alpha - Rz\|^2 + \sum_i \left\{ \psi_i(d_{\theta,i}) - d_{\theta,i}z_i + \frac{1}{2}H_{ii}z_i^2 \right\} \\ & + \mu^\top(b_\theta - A_\theta z) + \lambda_\theta \sum_j \left\{ \kappa_j|(Dz - h)_j| - \tau_j(Dz - h)_j \right\}. \end{aligned} \quad (34)$$

Thus the loss separates into resource-price error, coordinate-response error, unused priced continuation capacity, and switching-price error.

For a local statement, let  $z_0$  be the exact restricted optimizer at  $\theta_0$ , let the cached price attain equality there, and set  $v_0 = d_{\theta_0}(a) - Hz_0 \in N_{[\ell,u]}(z_0)$ . Suppose the query optimizer  $z_\theta$  stays on the priced face:  $\mu^\top(b_\theta - A_\theta z_\theta) = 0$ ,  $v_0 \in N_{[\ell,u]}(z_\theta)$ , and every switching term in (34) vanishes. With  $e_\theta = d_\theta(a) - Hz_\theta - v_0$ , one has

$$0 \leq \mathcal{U}_\theta(a) - K_\theta \leq \frac{1}{2} \|R(z_\theta - z_0)\|^2 + \frac{1}{2} e_\theta^\top H^{-1} e_\theta. \quad (35)$$

In particular, locally Lipschitz coefficients and a locally Lipschitz optimizer give a quadratic reuse loss on this priced face. The global upper certificate does not require these local conditions.

*Proof.* Linearize the resource square with  $\alpha$ , lower-bound each absolute value by its normalized tension, and subtract the feasible inequality and equality residuals. Maximizing the remaining separate concave quadratics on the box gives (33). Adding and subtracting their values at  $z$  gives (34); equality feasibility removes the equality-price residual. At an exact anchor the resource term implies  $\alpha = Rz_0$ , and each coordinate is a box maximizer, giving  $v_0 \in N_{[\ell,u]}(z_0)$ . On the priced face only the first two terms remain. For a box-feasible displacement  $y - z_\theta$ , the second term is the supremum of  $(v_0 + e_\theta)^\top(y - z_\theta) - \|y - z_\theta\|_H^2/2$ . The normal term is nonpositive; the unrestricted supremum of the remainder is  $e_\theta^\top H^{-1} e_\theta/2$ . This proves (35). EC Section EC.11 gives explicit constants and the boundary cases.  $\square$

The continuation prices need not be refitted to remain valid. They may, however, become expensive prices for slack that the query optimizer no longer uses. This distinction explains both the benefit of reuse and its failure mode. With affine coefficient dependence in  $d$  parameters, cached matrix-price products reduce evaluation to order  $n(d+1)$  arithmetic operations per anchor; exact rational bit cost and all anchor construction remain additional work. This is not the dimension-free reward-only bound of Proposition 7.1, and it is not a measured speedup.

The quadratic conclusion must not be extended across an arbitrary degenerate face. For  $|t| \leq 1/4$ , the problem  $\max\{x - x^2/2 : -1/2 \leq x \leq 1/2, x \leq t, x \leq -t\}$  has the following behavior. An exact price  $(1, 0)$  at  $t = 0$  gives  $\mathcal{U}_t = t$ , whereas  $K_t = -|t| - t^2/2$ . The reuse gap is  $2t + t^2/2$  for  $t > 0$ . The released first constraint has positive cached price and positive slack. Numerical anchor error likewise does not disappear by declaring an anchor exact: (33) remains valid, but the local quadratic assertion requires the exact-anchor assumptions.

**Corollary 7.4** (Uniform certification with correlated uncertainty). *Suppose all  $\theta$ -dependent coefficients in (31) are affine on a compact polytope  $\Theta$ , and  $\lambda_\theta \geq 0$  there. Partition  $\Theta$  into polytopes  $C$  with vertex sets  $V_C$ . A fixed implemented policy  $\hat{x}$  is feasible for the full accepted class throughout  $\Theta$  if it satisfies its affine inequalities and equalities at all vertices. For any finite cache  $\mathcal{A}$ ,*

$$\inf_{\theta \in \Theta} \{W_\theta(\hat{x}) - K_\theta\} \geq \min_C \max_{a \in \mathcal{A}} \min_{v \in V_C} \{W_v(\hat{x}) - \mathcal{U}_v(a)\}. \quad (36)$$

*Subdivision of cells cannot decrease this lower certificate, and adding cached prices cannot decrease it. Neither operation requires new comparator solves.*

*Proof.* The function  $\mathcal{U}_\theta(a)$  is convex in  $\theta$ : its residuals are affine and each  $\psi_i$  is convex. The fixed-policy reward is affine, so their difference is concave and is bounded below on a cell by its minimum vertex value. Choose the best fixed anchor on that cell, then take the worst cell. Convexity also proves the subdivision assertion; retaining old prices proves the cache assertion.  $\square$

The anchor must remain fixed inside the vertex minimum in (36). Taking the best anchor separately at vertices and interpolating those best values is generally invalid. The EC gives an exact failure example and a positive-gain example: four cells certify at least 0.04919921 uniformly against an optimized restricted contract under jointly changing reward, friction, and participation coefficients. No sampled parameter is substituted for a continuous uncertainty guarantee. The set is designed, not statistically calibrated.

## 8 Computational Evidence

### 8.1 Continuous-state guarantees without history enumeration

Table 1 reports the new construction in Section 6.1. Bounds cover every inherited tier and feasible root promise, not only evaluation points. In the four-period design, refining from 25 to 289 anchors per date–regime reduces the worst root bound from 0.113253 to 0.019567. The sixteen-period design stores 800 lower anchors rather than constructing a 65,535-node binary history tree; its coarser grid gives a wider 0.193576 bound. These figures report approximation guarantees and storage, not matched-time superiority or a uniform complexity rate. The six designs, complete rational witnesses, and all eleven negative controls are specified in EC Section EC.10.

### 8.2 Protocol and common-instance comparator hierarchy

We use two explicitly different families. The first takes the first 24 nominal contexts of the frozen R24 design: 63 history nodes, two service coordinates, six resource-reward coordinates, all continuation rows, and four shared capacities. A pooled action is an equal *amendment* to the existing heterogeneous tier. The fixed-amendment class is the singleton zero amendment because the exact root payment prevents a nonzero uniform change. Calling this an optimized constant actual tier would be incorrect.

The second family has 63 nodes, one tier, a common baseline of  $1/2$ , an equiprobable binary public regime, and Markov stage rewards, curvature, and conditional continuation caps. All five classes are feasible

and pool actual tiers. The exact root commitment fixes the optimized constant tier at  $1/2$  but leaves time-only schedules nontrivial. Every adjacent comparison uses the same primitive instance. We vary reward and switching parameters across 24 seeded contexts; EC Section EC.9 gives the integer and rational design. Integer probability weights multiply all conditional weights by 32, so its raw value unit differs from the legacy family. We report shares within each family, not a cardinal comparison between the two families.

Table 2 reports the optimized increments. Full-history amendments retain substantial value beyond two-period memory on the legacy family. In the public-Markov family most of the gain comes from observing the current regime; the additional memory and full-history gains are smaller and can vanish within the numerical bracket on particular instances. This distinction is informative: the total legacy gain is not presented as a uniquely contractual effect or attributed entirely to learning. Regime-only controls omit inherited tiers and promises even in the Markov family. The exact augmented-state result remains Theorem 6.1.

Each optimized class has a rational feasible policy and a separate rational dual upper bound. If these are  $[L_j, U_j]$ , the certified interval for an increment is  $[\max\{0, L_j - U_{j-1}\}, U_j - L_{j-1}]$ . We do not interpret a tiny difference between two approximate lower endpoints as a negative true increment. Both the lower-endpoint summaries and these enclosing intervals are retained in the raw record. Exact symbolic tests independently verify the three-node release path, the capacity rent example, and the optimized friction example. Additional rational matrix tests verify the regular-cell release formula. Finite-action promise recursions match exhaustive history enumeration at every attainable root promise; these finite checks supplement the induction proof rather than replace it.

### 8.3 Radius refinement and comparator reuse

Table 3 recomputes the true and common-outer time-only comparators on eight fixed legacy contexts and ten radii, including much smaller positive radii than the earlier study. Each hidden coefficient vector is held fixed while its radius varies. The outer-set slack tends toward zero and rises steeply before the earlier smallest positive radius. Thus the coarse positive-radius plateau did not establish a discontinuity. The construction already depended continuously on radius; Proposition 7.2 explains its local behavior. This experiment isolates the outer-set contribution and does not conflate it with objective interpolation over uncertainty vertices.

For comparator reuse we fix the legacy time-only geometry and switching coefficient, build four certified anchors, and validate 32 nearby reward queries. The rational trace curvature bound is approximately 0.45830. The largest envelope-minus-certified-lower-value gap is reported in Table 4. The deployed envelope makes no query optimization calls; the 32 validation solves are offline work used to measure tightness. The four initial solves and storage are not free. This is a test of a reusable upper bound and its quality, not a measured end-to-end speedup over the historical robust pipeline.

### 8.4 Correlated comparator certification without query optimization

Table 5 evaluates Theorem 7.3 on the same joint coefficient family as the outer-comparator study, rather than weakening the accepted constraints. Three historical prices per context form a frozen cache. Its first panel

contains the 56 non-anchor queries from the finer-radius record. Its second panel contains 24 new parameter vectors outside those inherited one-dimensional rays. A separate true restricted solve, outer solve, and full accepted solve are used only for offline evaluation at each new query. The deployed cache receives none of their optimizers or values.

Across the 24 new queries, the mean certified upper slack decreases from 0.064069 for the historical outer comparator to 0.016632 for the three-price cache, with both figures rounded upward. All 24 cache bounds are tighter than their matched outer bounds. The implemented full-class policies have positive cached economic-gain lower bounds in every case, with minimum at least 0.762195 in this family’s objective units. Every optimization bracket is below  $6 \times 10^{-8}$ . These are finite, designed comparisons, not a distributional performance or a field-calibration result.

A finite cache is not automatically tight away from its anchors. The one-price column worsens at larger perturbations; adding prices helps without compromising validity. Nor do these calculations establish an end-to-end speed advantage: exact rational arithmetic, all initial solves, the implemented-policy audit, and any requested cache refresh must still be charged. The result is that a new restricted optimization is not logically required to produce the displayed bound. EC Section EC.11.4 specifies all new coefficients, witnesses, source inheritance, and exact checks. Corollary 7.4 separately provides a continuous uncertainty guarantee; the pointwise table is not substituted for that proof.

## 8.5 Historical learning and timing evidence

The R24 matched study charged optimizer labels, fitting, prediction, auditing, polishing, and fallback against lifted continuation. At the strict target both learned routes fell back in every measured case at 63, 127, and 255 nodes. Their mean total times exceeded the lifted comparator in every configuration, with unfavorable paired intervals. The coarse target also produced no observed finite amortization case. Table 6 retains the strict-target comparisons; the complete phase-resolved record and all less favorable and more favorable individual cases remain in the code/data archive. These are implementation diagnostics, not a second positive contribution about learned optimization.

Let  $C_{\text{off}} \geq 0$  be a candidate’s incremental offline cost, and let  $t_C, t_L$  be its and the lifted method’s per-query times. The candidate-minus-lifted total cost after  $Q_0$  queries is

$$C_{\text{off}} + Q_0(t_C - t_L) = C_{\text{off}} - Q_0\delta, \quad \delta = t_L - t_C. \quad (37)$$

For positive offline cost a finite positive crossing exists only when  $t_C < t_L$ , at  $Q_0 = C_{\text{off}}/(t_L - t_C)$ . Strict savings require a query count above this crossing. The analysis code uses this convention, and the displayed equation now matches it. The historical observations have no positive learned crossing under that protocol.

The historical robust study compared different achieved gains. Its roughly 20-millisecond differences between protected learned policies and robust maximin are therefore descriptive cost accounting, not matched-quality dominance. We do not use these three isolated method points as a computational advantage claim. Their complete gain–cost coordinates remain in the archive. The reward-only envelope applies under explicitly fixed geometry, while Theorem 7.3 retains jointly changing coefficients. Neither is retroactively

substituted into the old timings; their certificate quality is assessed separately.

## 9 Conclusions

An accepted adaptation premium is a difference between optimized contract values. The release frontier identifies the gain from relaxing an information restriction, its curvature, and the first continuation slack that limits the improvement. The comparator-adjusted rent decomposition explains why a capacity relaxation can lower that premium and why greater friction can increase it. These statements depend on the actual restriction, not on a presumption that more capacity or less friction has the same effect on both optimized classes.

The public-state institution also separates contractual memory from the exogenous regime. A remaining-payment promise and inherited tier can make a history policy recursively implementable, but the resulting continuous state still carries a computational cost. The numerical hierarchy shows which restrictions matter in each designed family instead of assigning all positive value to a convenient time-only benchmark. Independent policy and comparator certificates then connect these structural values to implemented decisions. Learning remains one possible implementation route; the tested protocol provides no evidence that it is the cheapest one.

When operating conditions change together, the comparator need not be repeatedly replaced by a loose outer model. Cached normalized prices preserve the actual continuation constraints. Their gap decomposition identifies the price of reuse, while a cellwise certificate extends the comparison from known queries to a designed uncertainty set. The resulting positive-gain statement is separate from, and does not depend on, a claim that a particular predictor is faster.

## Data and Code Accessibility

The separate code/data package contains the R25 source, rational certificate records, exact example checks, finite-action recursion checks, replay and build scripts, a point-by-point response, and a preservation manifest. It also contains the complete predecessor manuscript, formal companion, computational record, historical supplement, and earlier repository derivations. None of the predecessor mathematical statements or empirical tables is silently discarded. The current manuscript and its formal electronic companion form the journal-facing argument; the predecessor and computational documents are an indexed research archive rather than additional required theorem layers. No field data are used. Repository delivery is for further author/referee review and is not a submission to the journal.

The R26 directory adds the continuous-state generators, independent rational verifier, point-by-point response to both R24 reports, and byte-identical R25 predecessor copies. The inherited R25/R24 results are replayed rather than relabeled as new measurements. Both formal documents retain their predecessor mathematical statements and proofs; the new continuous-state and omitted-information results are integrated in Section 6.

The R27 directory adds the correlated-price theorem, its uniform-uncertainty extension, 72 new opti-

mization proposals, independent rational checks, a query-only cache interface, and a complete response to both latest reports. It preserves the R26 root files byte for byte and hashes every inherited file. The new evaluations of old radius records are distinguished from new off-ray solves.

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Table 1: Continuous promised-state certificates on designed Markov models

| Dates | Subdivisions | Lower anchors | Upper planes | Uniform gap | Tree nodes |
|-------|--------------|---------------|--------------|-------------|------------|
| 3     | 2            | 54            | 34           | 0.299587    | 7          |
| 4     | 4            | 200           | 77           | 0.113253    | 15         |
| 4     | 8            | 648           | 153          | 0.051939    | 15         |
| 4     | 16           | 2,312         | 322          | 0.019567    | 15         |
| 8     | 4            | 400           | 175          | 0.136908    | 255        |
| 16    | 4            | 800           | 362          | 0.193576    | 65,535     |

Lower anchors and upper planes are totals over all dates and both regimes. The uniform gap is the largest certified root loss over both regimes, all inherited tiers, and all feasible promises, rounded upward to six decimals. Tree nodes give the hypothetical full binary history tree, which is not constructed. These rows are not timing comparisons.

Table 2: Value of progressively richer optimized contracts

| Expansion                         | Legacy amendments |           | Markov service tiers |           |
|-----------------------------------|-------------------|-----------|----------------------|-----------|
|                                   | Mean gain         | Share (%) | Mean gain            | Share (%) |
| Fixed to time-only                | 0.116155          | 7.95      | 0.127557             | 0.49      |
| Time-only to regime               | 0.136312          | 9.33      | 26.066328            | 99.34     |
| Regime to two-period memory       | 0.224350          | 15.36     | 0.028788             | 0.11      |
| Two-period memory to full history | 0.983916          | 67.36     | 0.016938             | 0.06      |
| Total                             | 1.460733          | 100.00    | 26.239611            | 100.00    |

There are 24 common instances per family and five optimized classes per instance. Means use certified lower endpoints; the complete enclosing increment intervals are in the data. Maximum optimization-bracket widths are  $8.13\text{e-}08$  and  $2.97\text{e-}07$ , respectively. Shares divide each mean increment by the within-family total. Raw value units differ between families. Current regime omits inherited tier and remaining promise.

Table 3: The common outer comparator becomes locally tight

| Uncertainty radius | Mean upper bound on outer-set slack |
|--------------------|-------------------------------------|
| 0                  | 0.000000003                         |
| 1/65536            | 0.000047365                         |
| 1/16384            | 0.000189415                         |
| 1/4096             | 0.000757072                         |
| 1/1024             | 0.003019002                         |
| 1/256              | 0.011836705                         |
| 1/64               | 0.045064142                         |
| 1/16               | 0.073768497                         |
| 1/8                | 0.074890218                         |
| 1/4                | 0.078860565                         |

Eight fixed contexts and ten radii, with a directly optimized true and outer comparator at each pair (160 certificates). Entries average the outer certified upper value minus the true certified lower value. At zero the true outer-set slack is exactly zero; the displayed residual is the optimization bracket. Maximum individual bracket width is  $3.06\text{e-}08$ . This isolates outer-set loss and is not a calibrated uncertainty experiment.

Table 4: Restricted-comparator reuse with fixed geometry

| Quantity                                              | Value       |
|-------------------------------------------------------|-------------|
| Certified anchors                                     | 4           |
| Nearby reward queries                                 | 32          |
| Restricted optimizations in deployed query evaluation | 0           |
| Query optimizations used only for offline validation  | 32          |
| Curvature trace bound                                 | 0.458298401 |
| Mean envelope slack upper bound                       | 0.000129719 |
| Maximum envelope slack upper bound                    | 0.000248488 |

A common time-only feasible set and switching coefficient are held fixed. Each query evaluates all four analytical anchor envelopes and takes their minimum. The slack is cached upper value minus independently certified query lower value; it therefore encloses the actual upper-bound loss. Anchor construction, policy acceptance checks, storage, and offline validation are separate costs. No end-to-end timing advantage is inferred.

Table 5: Certified comparator upper slack under correlated parameter changes

| Radius                                               | Queries | Outer      | One price  | Three prices |
|------------------------------------------------------|---------|------------|------------|--------------|
| <i>Inherited ray holdouts; no query price reused</i> |         |            |            |              |
| 1/65536                                              | 8       | 0.00004737 | 0.00000001 | 0.00000001   |
| 1/16384                                              | 8       | 0.00018942 | 0.00000001 | 0.00000001   |
| 1/4096                                               | 8       | 0.00075708 | 0.00000011 | 0.00000011   |
| 1/1024                                               | 8       | 0.00301901 | 0.00000169 | 0.00000169   |
| 1/256                                                | 8       | 0.01183671 | 0.00002694 | 0.00002694   |
| 1/64                                                 | 8       | 0.04506415 | 0.00043085 | 0.00043085   |
| 1/8                                                  | 8       | 0.07489022 | 0.02778576 | 0.00711323   |
| <i>New off-ray queries; same frozen cache</i>        |         |            |            |              |
| 1/64                                                 | 8       | 0.04463703 | 0.00073960 | 0.00073960   |
| 1/16                                                 | 8       | 0.07337423 | 0.01428873 | 0.01428873   |
| 1/8                                                  | 8       | 0.07419502 | 0.05643124 | 0.03486551   |

Entries are means of  $U - L_K$ , where  $L_K$  is an independently checked restricted feasible value. Each decimal is rounded upward. The one-price cache uses the nominal anchor; the three-price cache also uses the inherited radii 1/16 and 1/4. No anchor query is included in the first panel. Off-ray queries change coefficient directions, not merely radii. Initial prices and the offline validation optimizations are charged separately; these are certificate-quality comparisons, not matched timings.

Table 6: Preserved strict-target end-to-end timing comparisons

| Nodes | Lifted (ms) | Tanh (ms) | RBF (ms) |
|-------|-------------|-----------|----------|
| 63    | 10.34       | 18.84     | 18.91    |
| 127   | 96.24       | 112.53    | 113.62   |
| 255   | 239.93      | 277.86    | 274.96   |

Historical R24 measurements, unchanged. Target  $10^{-5}$ ; 32 held-out contexts and three repetitions per method and size. Both learned routes have 100% fallback at each size. Their paired saving intervals versus lifted continuation are entirely negative. Offline label/fitting costs are retained separately and cannot turn negative online savings into amortization. These are not newly timed R25 measurements.